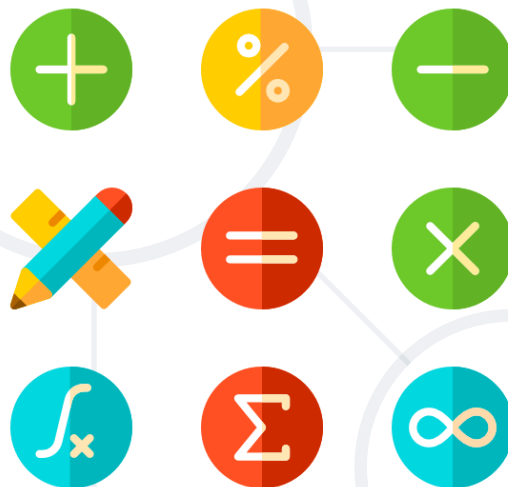


Built-in Functions

Functions and Wildcards in SQL Server



SoftUni Team
Technical Trainers



SoftUni



Software University

<https://softuni.bg>

Table of Contents

1. Function Overview
2. String Functions
3. Math Functions
4. Date Functions
5. Other Useful Functions
6. Wildcards





Functions in SQL Server

Overview

- **Aggregate functions**

- Perform a calculation on a set of values and return a single value
- Examples: **AVG, COUNT, MIN, MAX, SUM**

- **Analytic functions**

- Compute an aggregate value based on a group of rows
- Unlike aggregate functions, analytic functions can return multiple rows for each group

```
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY Salary DESC)  
OVER (PARTITION BY DepartmentId) AS MedianCont
```



SQL Functions



- **Ranking functions**

- Return a ranking value for each row in a partition
- **RANK, ROW_NUMBER, DENSE_RANK, NTILE (OVER)**

- **Rowset functions**

- Return an object that can be used like table references in a statement
- **OPENDATASOURCE, OPENJSON, OPENXML, OPENROWSET**

- **Scalar functions**

- Operate on a single value and then return a single value
- Scalar functions can be used wherever an expression is valid



String Functions

- **Concatenation** – combines strings

```
SELECT FirstName + ' ' + LastName  
       AS [Full Name]  
FROM Employee
```

```
SELECT CONCAT(FirstName, ' ', LastName)  
       AS [Full Name]  
FROM Employee
```

- **CONCAT** replaces **NULL** values with **empty string**
- **CONCAT_WS** combines strings with separator

- **SUBSTRING** – extracts a part of a string

```
SUBSTRING(String, StartIndex, Length)
```

```
SUBSTRING('SoftUni', 5, 3)
```



Uni

- Example: get short **summary** of an article

```
SELECT ArticleId, Author, Content,  
       SUBSTRING(Content, 1, 200) + '...' AS Summary  
FROM Articles
```

- **REPLACE** – replaces a specific string with another

```
REPLACE(String, Pattern, Replacement)
```

```
REPLACE('SoftUni', 'Soft', 'Hard')
```



```
HardUni
```

- Example: **cancel** the word blood from album names

```
SELECT REPLACE(Title, 'blood', '*****')  
  AS Title  
FROM Album
```

- **LTRIM** & **RTRIM** – remove spaces from either side of string

```
LTRIM(String)
```

```
RTRIM(String)
```

- **LEN** – counts the number of characters

```
LEN(String)
```

- **DATALENGTH** – gets the number of used bytes

```
DATALENGTH(String)
```

- **LEFT & RIGHT** – get characters from the beginning or the end of a string

```
LEFT(String, Count)
```

```
RIGHT(String, Count)
```

- Example: name **shortened** (first 3 letters)

```
SELECT Id, Start,  
       LEFT(Name, 3) AS Shortened  
FROM Games
```

- **LOWER & UPPER** – change letter casing

```
LOWER(String)
```

```
UPPER(String)
```

- **REVERSE** – reverses order of all characters in a string

```
REVERSE(String)
```

- **REPLICATE** – repeats a string

```
REPLICATE(String, Count)
```

- **FORMAT** – format a value with a valid .NET format string

```
FORMAT(SomeDate, 'yyyy-MMMM-dd', 'bg-BG')
```

Problem: Obfuscate CC Numbers

- The database contains credit card details for customers
- Provide a summary without revealing the serial numbers

ID	FirstName	LastName	PaymentNumber
1	Guy	Gilbert	5645322227179083
2	Kevin	Brown	4417937746396076
...	...		...



ID	FirstName	LastName	PaymentNumber
1	Guy	Gilbert	564532*****
2	Kevin	Brown	441793*****
...	...		...

Solution : Obfuscate CC Numbers

- Reveal the first 6 digits and obfuscate the rest

```
SELECT CustomerID,  
       FirstName,  
       LastName,  
       LEFT(PaymentNumber, 6) + '*****'  
FROM Customers
```

- Bonus – create a View for the use of clients

```
CREATE VIEW v_PublicPaymentInfo AS
```

```
...
```

- **CHARINDEX** – locates a specific pattern (substring) in a string

Optional, begins at 1

CHARINDEX(Pattern, String, [StartIndex])

- **STUFF** – inserts a substring at a specific position

STUFF(String, StartIndex, Length, Substring)

Number of chars
to delete



Math Functions

Arithmetic, PI, ABS, ROUND, Etc.

- 

Id	Area
1	4
2	9
3	6.75
4	48
5	7.5

[illegible]

- **PI** – gets the value of Pi as a float (15 –digit precision)

```
SELECT PI() --3.14159265358979
```

- **ABS** – absolute value

```
ABS(Value)
```

- **SQRT** – square root (the result will be float)

```
SQRT(Value)
```

- **SQUARE** – raise to power of two

```
SQUARE(Value)
```

Example: Line Length

- Find the length of a line by given coordinates of the end points

Id	X1	Y1	X2	Y2
1	0	0	10	0
2	0	0	5	3
4	-1	5	8	-3
5	18	23	8882	134



Id	Length
1	10
2	5.8309518948453
4	12.0415945787923
5	8864.69497501183

```
SELECT Id,  
       SQRT(SQUARE(X1-X2) + SQUARE(Y1-Y2))  
       AS Length  
FROM Lines
```

- **POWER** – raises value to the desired exponent

```
POWER(Value, Exponent)
```

- **ROUND** – obtains the desired precision

- Negative precision rounds characters before the decimal point

```
ROUND(Value, Precision)
```

- **FLOOR & CEILING** – return the nearest integer

```
FLOOR(Value)
```

```
CEILING(Value)
```

- Calculate the required number of pallets to ship each item
 - **BoxCapacity** specifies how many items can fit in one box
 - **PalletCapacity** specifies how many boxes can fit in a pallet

Id	Name	Quantity	BoxCapacity	PalletCapacity
1	Perlenbacher 500ml	108	6	18
2	Perlenbacher 500ml	10	6	18
3	Chocolate Chips	350	24	3
4	Oil Pump	100	1	12
5	OLED TV 50-Inch	13	1	5
6	Penny	1	2239488	1



Number of pallets
1
1
5
9
3
1

- Since we can't use half a box or half a pallet, we need to round up to the nearest integer value

```
SELECT
    CEILING(
        CEILING(
            CAST(Quantity AS float) /
            BoxCapacity) / PalletCapacity)
    AS [Number of pallets]
FROM Products
```

- **SIGN** – returns 1, -1 or 0, depending on the value of the sign

```
SIGN(Value)
```

- **RAND** – gets a random float value in the range [0, 1]
 - If Seed is not specified, it will be assigned randomly

```
RAND()
```

```
RAND(Seed)
```



Date Functions

GETDATE, DATEDIFF, DATEPART, Etc.

- **DATEPART** – extract a segment from a date as an integer
 - Part can be any part and format of date or time

DATEPART(Part, Date)

year, yyyy, yy

month, mm, m

day, dd, d

YEAR(Date)

MONTH(Date)

DAY(Date)

- For a full list, take a look at the [official documentation](#)

Problem: Quarterly Report

- Prepare sales data for aggregation by displaying yearly quarter, month, year and day of sale

InvoiceId	InvoiceDate	Total
1	2023-01-01	1.98
2	2023-01-02	3.96
3	2023-01-03	5.94
4	2023-01-06	8.91



InvoiceId	Total	Quarter	Month	Year	Day
1	1.98	1	1	2023	1
2	3.96	1	1	2023	2
3	5.94	1	1	2023	3
4	8.91	1	1	2023	6

- Use **DATEPART** to get the relevant parts of the date

```
SELECT InvoiceId, Total,  
       DATEPART(QUARTER, InvoiceDate) AS Quarter,  
       DATEPART(MONTH, InvoiceDate) AS Month,  
       DATEPART(YEAR, InvoiceDate) AS Year,  
       DATEPART(DAY, InvoiceDate) AS Day  
FROM Invoice
```

- This statement might be useful as a View

- **DATEDIFF** – finds the difference between two dates
 - **Part** can be **any part** and **format** of date or time

```
DATEDIFF(Part, FirstDate, SecondDate)
```

- Example: Show employee experience

```
SELECT ID, FirstName, LastName,  
       DATEDIFF(YEAR, HireDate, '2017/01/25')  
       AS [Years In Service]  
FROM Employees
```

- **DATENAME** – gets a string representation of a date's part

```
DATENAME(Part, Date)
```

```
SELECT DATENAME(weekday, '2017/01/27')
```

- **DATEADD** – performs date arithmetic
 - **Part** can be **any part** and **format** of date or time

```
DATEADD(Part, Number, Date)
```

- **GETDATE** – obtains the current date and time

```
SELECT GETDATE()
```

- **EOMONTH** – returns the last day of the month



Other Functions

CAST, CONVERT, OFFSET, FETCH

- **CAST & CONVERT** – conversion between data types

```
CAST(Data AS NewType)
```

```
CONVERT(NewType, Data)
```

- **ISNULL** – swaps **NULL** values with a specified **default value**

```
ISNULL(Data, DefaultValue)
```

- Example: Display "Not Finished" for projects with no **EndDate**

```
SELECT ProjectID, Name,  
       ISNULL(CAST(EndDate AS varchar), 'Not Finished')  
FROM Projects
```

- **COALESCE** – evaluates the arguments in order and returns the current value of the first expression that initially does not evaluate to **NULL**

```
SELECT COALESCE(NULL, NULL, 'third_value',  
  'fourth_value');  
  
// third_value
```

- **OFFSET & FETCH** – get only specific rows from the result set
 - Used in combination with **ORDER BY** for pagination

```
SELECT ID, FirstName, LastName  
FROM Employees  
ORDER BY ID  
OFFSET 10 ROWS  
FETCH NEXT 5 ROWS ONLY
```

Rows to skip

Rows to include

- **ROW_NUMBER** – always generate unique values without any gaps, even if there are ties
- **RANK** – can have gaps in its sequence and when values are the same, they get the same rank
- **DENSE_RANK** – returns the same rank for ties, but it doesn't have any gaps in the sequence
- **NTILE** – Distributes the rows in an ordered partition into a specified number of groups



Wildcards

Selecting Results by Partial Match

- **Wildcards** are used with **WHERE** for partial filtration
- Similar to **Regular Expressions**, but **less capable**
- Example: Find all employees who's first name **starts with "Ro"**

```
SELECT ID, FirstName, LastName  
FROM Employees  
WHERE FirstName LIKE 'Ro%'
```

Wildcard symbol

- Supported characters include:

```
%      -- any string, including zero-length
_      -- any single character
[... ] -- any character within range
[^... ] -- any character not in the range
```

- ESCAPE** – specify a prefix to treat special characters as normal

```
SELECT ID, Name
FROM Tracks
WHERE Name LIKE '%max!%' ESCAPE '!'
```

- Various **built-in functions**
- String functions - **CONCAT, LEFT / RIGHT, REPLACE**, etc.
- Math functions - **PI, ABS, POWER, ROUND**, etc.
- Date functions - **DATEPART, DATEDIFF, GETDATE**, etc.
- Using **Wildcards**, we can obtain results by partial string matches



SoftUni Diamond Partners



**SUPER
HOSTING
.BG**

encorp.ai

createX

INDEAVR
Serving the high achievers


**DRAFT
KINGS**

THE CROWN IS YOURS

VIVACOM

SoftUni Diamond Partners



THE CROWN IS YOURS



- Software University – High-Quality Education, Profession and Job for Software Developers

- softuni.bg, about.softuni.bg

- Software University Foundation

- softuni.foundation

- Software University @ Facebook

- facebook.com/SoftwareUniversity



- This course (slides, examples, demos, exercises, homework, documents, videos and other assets) is **copyrighted content**
- Unauthorized copy, reproduction or use is illegal
- © SoftUni – <https://about.softuni.bg>
- © Software University – <https://softuni.bg>

